

Fixing India's air pollution problem

MAKING AIR BREATHABLE ONE STEP AT A TIME!

Jayesh Shinde | jayesh@digit.in

In 2022, India had 12 of the 15 most polluted cities in terms of air in the world, according to

IQAir – a Swiss monitoring agency's annual report. One has to wonder if there's any corner of the country where the air isn't potentially hazardous to our lungs.

Against such insurmountable odds, Angad Daryani's Praan is helping solve the air pollution crisis through huge 20-foot high industrial air purifiers, the Praan MKII. An inventor and social entrepreneur with an electrical engineering degree from the Georgia Institute of Technology, Daryani's moonshot with Praan has been his ability to engineer a filterless air purifier design – which subsequently refined to its current version, the MKII.

In an email interview, Daryani told us more about the MKII and how it's fighting air pollution not just in India but around the world.

Q Can you elaborate on the technological innovation behind the MKII that allows it to be a truly "filterless" air purification device?

A MKII is based on charging pollutant particles, separating them hyperlocally and collecting them in a collection chamber. What makes our technology unique is not just the ability to capture particulates without filters, but doing so at high flow rates, and being effective with conductive dust, outlier dust like limestone, as well as sticky dust.

Q How does the performance of the MKII compare to traditional

filtered air purifiers in terms of particulate matter and carbon capture?

A MKII is just a filterless particulate matter removal system. It's the highest rate of removal of any filterless system - and the only one effective with conductive dust, sticky dust, and outlier dust like limestone. It's also the one which allows reuse of all captured dust on a daily basis without changing the chemical constituency of the dust - very useful for large dirty industries like steel/cement/refractories/etc.

Q Given its filterless design, how is the maintenance of the MKII managed? What about its durability and lifespan compared to traditional air purifiers?

A MKII has a 5-7 years base lifespan and requires service and recalibration once a year. This is performed on site for the customers. Other systems are ESPs and bag filters - both require expensive main-

tenance, ducting systems, and part replacement every year. Please see the comparison with other systems below:

Q How adaptable is the MKII to different environments (e.g., residential, commercial, industrial)? Can it be easily scaled depending on the requirements of the space?

A We have primarily designed and made it for Industrial spaces – however, it can also be used in airports, train stations, etc – renders below. The device is literally plug and play.

Q How energy-efficient is MKII? In terms of overall sustainability and environmental impact, what measures have been taken?

A As a climate-tech company, energy efficiency is essential to us. The device which is manufactured is entirely net-zero with no parts which need to be changed on an operational basis. All emissions which went into the production of these devices are



MKII being installed or transported